

AURA

Customer Ordering System

CSE323 · Spring 2026 · Flask · Firebase · SQLite

Phase 1

Requirements

Phase 2

Design

Phase 3

Implementation

Phase 4

Validation

PHASE 1

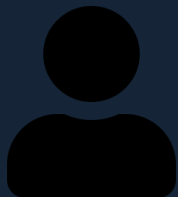
Requirement Discovery

Actor Classification

Traceability Heatmap

Persona Edge Cases

Actors



Customer

role: customer

PRIMARY



Admin

role: admin

SUPPORTING



Delivery

role: delivery

SUPPORTING



Firebase

Identity Toolkit

OFFSTAGE

Traceability

35

Requirements

0

Orphaned

3

Gaps Disclosed

8

REQ Categories

AUTH
7

CHK
5

PRO
5

ADM
3

MENU
3

CART
3

Requirements by category

8 Edge Cases Found

via AI Persona Avatar



Typo Referral Code

400



Stock Depleted at Checkout

FAIL



Cart > Stock

FAIL



Cancel After PREPARING

ERROR



Unverified Email Login

401



Advance Wrong Status

FAIL



Promo Below Min Order

REJECT



Expired Voucher

EXPIRED

PHASE 2

Design & Specification

Gherkin Scenarios

UML Diagrams

API Contracts

QA Refinement

2

Gherkin Scenarios

1 Feature: Customer Registration

3 Scenario: Successful registration

4 Give Firebase is configured

5 When POST with valid fields

6 Then User row created in DB

8 Scenario: Promo below min order

9 Give Subtotal < \$30

10 When Validate AURA20 promo

11 Then Error: min order message



Auth



Cart



Checkout



Admin

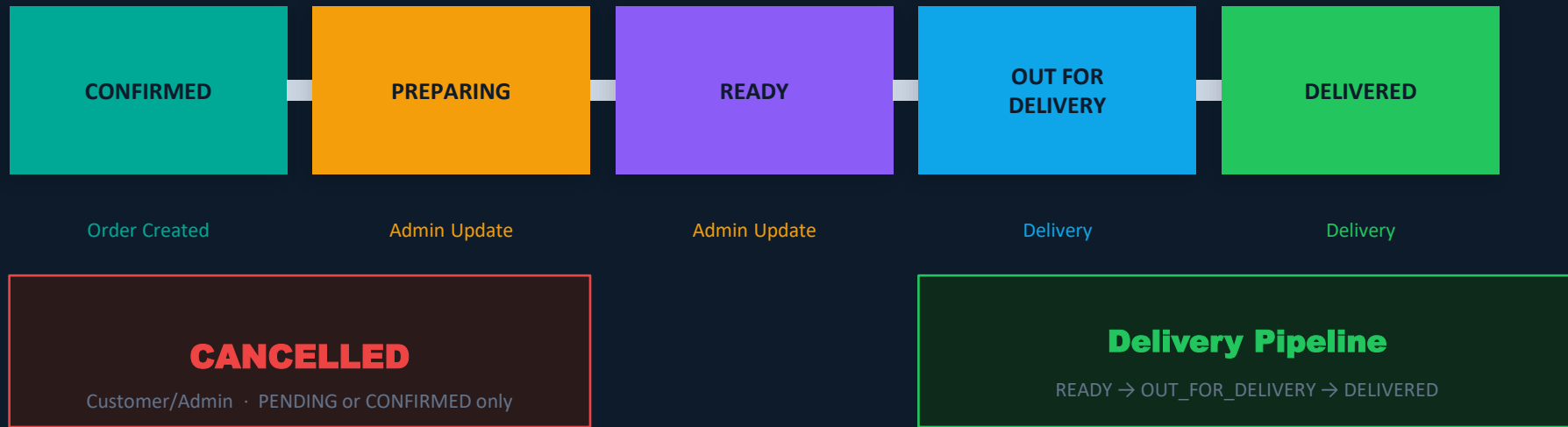


Delivery



Vouchers

Order Lifecycle



QA Refinement

✗ VAGUE

"Secure Login"



"Valid Card"



"Active Voucher"



"Minimum Order"



✓ MEASURABLE

Werkzeug hash (staff)
Firebase + emailVerified (customer)
UID mismatch → 401

Luhn algorithm pass
Length 13–19 digits
CVV 3–4 digits · expiry not past

is_used == False
AND utcnow() < expires_at

subtotal ≥ promo.min_order
before discount applied

PHASE 3

Test-Driven Implementation

8 Sprints · Vertical Slices

TDP: Failing → Padlock → Iterate

AI Prompts Disclosed



8 Sprints

1



Auth & RBAC

2



Menu & Cart

3



Checkout

4



Admin Panel

5



Delivery

6



Order Tracking

7



Profile & Rewards

8



Static Pages

Edge Case Cage



Write
Failing Test



Add
Padlocks



Prompt
AI



Iterate to
Passing



Rating 1–5

inclusive



Password

≥ 8 chars



Card CVV

3 or 4 digits



Cart Qty

$\leq$ stock



Points Redeem

≥ 500



Promo Min

subtotal $\geq$ min



Delivery Advance

READY or OUT



Referral Reward

1st DELIVERED

PHASE 4

Validation & Pipeline

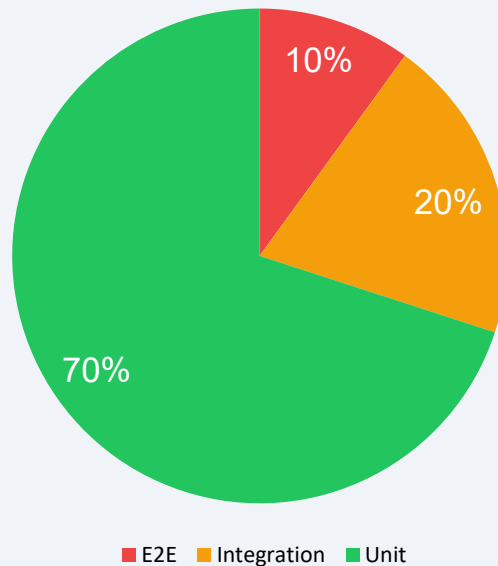
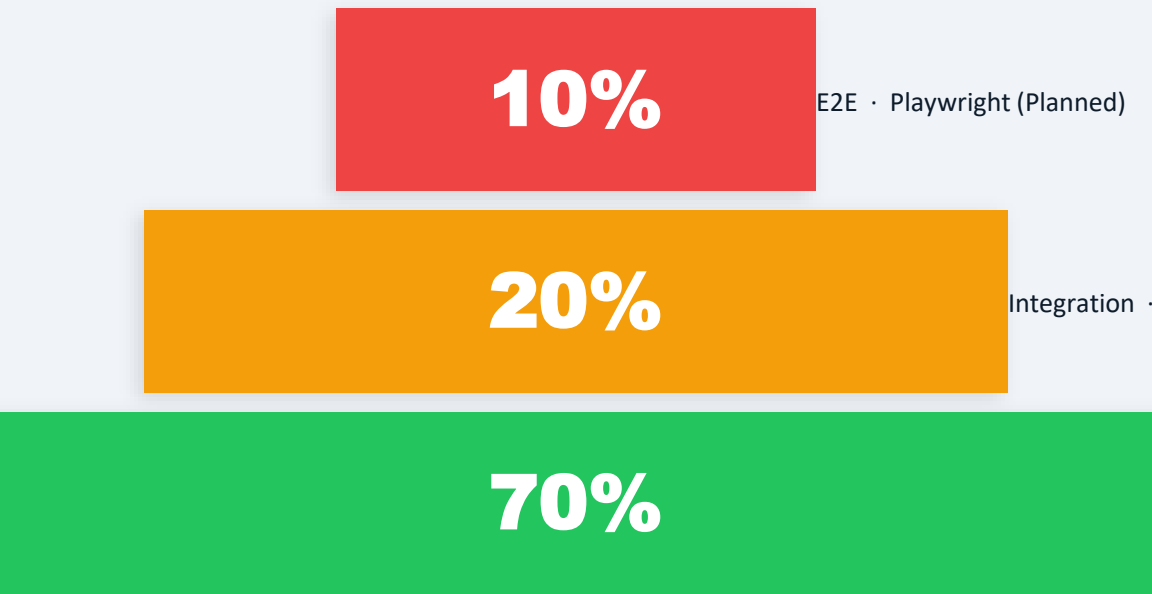
70% Unit · 20% Integration · 10% E2E

10 Test Modules

Playwright POM (Planned)



Testing Pyramid



10 Test Modules · `pytest -v --cov=app`

Verified vs Validated



VERIFICATION

Did we build
it RIGHT?

-
- ✓ Routes → correct templates/JSON
 - ✓ RBAC → 403/401 enforced
 - ✓ Luhn + CVV validation
 - ✓ Services enforce business rules



VALIDATION

Did we build the
RIGHT THING?

-
- ✓ Browse → Cart → Checkout
 - ✓ Admin: PREPARING → READY
 - ✓ Delivery: OUT → DELIVERED
 - ✓ Customer receives loyalty points



GitHub Repository

github.com/mona585/Customer-Ordering-System

Website link

<https://customer-ordering-system.onrender.com>

